Derivatives: Contracts, Models, and Markets · Technical Note 28

The Fundamental Theorems of Asset Pricing

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Chapter 28 of Derivatives: Contracts, Models, and Markets uses two results without proving them: that no arbitrage is equivalent to the existence of an equivalent martingale measure, and that completeness is equivalent to that measure being unique. This note states both precisely, proves them where the proof is elementary, and says what has to change when the market runs in continuous time.

The order of business is deliberate. A market with finitely many states is where both theorems are exactly true and where the reason is visible, because everything reduces to linear algebra. Continuous time then needs a repair that is not a technicality, and Section 3 is about that repair.

1 THE FINITE MARKET

Fix one period. There are n states of the world at date 1 and k traded assets. Asset j costs S_j today and pays D_{ji} in state i, so the payoff matrix is $D \in \mathbb{R}^{k\times n}$ and the price vector is $S \in \mathbb{R}^k$.

A **portfolio** is a vector $\theta \in \mathbb{R}^k$, costing $\theta^\top S$ today and paying $\theta^\top D$ across states. An **arbitrage** is a portfolio that is never worse than nothing and sometimes better:

$$\theta^\top S \le 0, \qquad \theta^\top D \ge 0, \qquad \text{with at least one inequality strict.} \tag{TN28.1}$$

A **state-price vector** is a strictly positive $\psi \in \mathbb{R}^n_{++}$ with

$$S = D\psi \tag{TN28.2}$$

so that every asset's price is the sum across states of its payoff times the price of a unit paid in that state.

State prices and martingale measures are the same object seen twice. Given ψ, write $\psi_0 = \sum_i \psi_i$ for the price today of one unit in every state (a riskless bond), and set

$$q_i = \frac{\psi_i}{\psi_0}, \qquad 1 + r = \frac{1}{\psi_0} \tag{TN28.3}$$

The q_i are strictly positive and sum to one, so they are a probability measure; and $\psi_i > 0$ for every i is exactly the statement that Q is equivalent to the physical measure, since neither assigns zero probability to any state. Rewriting (TN28.2) with (TN28.3) gives

$$S_j = \frac{1}{1+r}\sum_{i=1}^{n} q_i D_{ji} = \frac{1}{1+r} E^Q[D_j] \tag{TN28.4}$$

which is the martingale property of Chapter 28's (28.3) in one period.

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The first theorem

> **First fundamental theorem.** There is no arbitrage if and only if a strictly positive state-price vector exists or, equivalently, an equivalent martingale measure exists.

One direction is immediate. Suppose $\psi > 0$ solves $S = D\psi$ and let θ be any portfolio. Then

$$\theta^\top S = \theta^\top D\psi \tag{TN28.5}$$

If $\theta^\top D \geq 0$ with some entry strictly positive, the right-hand side is strictly positive because every ψ_i is, so $\theta^\top S > 0$ and (TN28.1) cannot hold. An arbitrage is impossible.

The converse is the substantive half, and it is a separating-hyperplane argument. Consider the set of payoff-and-cost pairs a portfolio can attain,

$$M = \left\{\left(-\theta^\top S,\ \theta^\top D\right) : \theta \in \mathbb{R}^k\right\} \subseteq \mathbb{R}^{1+n}$$

a linear subspace, and the nonnegative cone $K = \mathbb{R}^{1+n}_+ \setminus \{0\}$. No arbitrage says precisely that $M \cap K = \emptyset$. A subspace disjoint from the nonnegative cone can be separated from the simplex $\{x \in \mathbb{R}^{1+n}_+ : \sum_a x_a = 1\}$, which is compact and convex, by a vector $\phi \in \mathbb{R}^{1+n}$ with $\phi^\top x > 0$ on the simplex and $\phi^\top m = 0$ for all $m \in M$. The first condition forces every entry of ϕ to be strictly positive; the second, written out, is

$$-\phi_0\, \theta^\top S + \sum_{i=1}^{n} \phi_i\, (\theta^\top D)_i = 0 \qquad \text{for every } \theta$$

Dividing by $\phi_0 > 0$ and setting $\psi_i = \phi_i/\phi_0$ gives $\theta^\top S = \theta^\top D\psi$ for all θ, hence $S = D\psi$ with $\psi > 0$. ■

Nothing in that argument is probabilistic. It is the duality between the absence of arbitrage and the existence of a positive linear pricing rule, and the probability measure only appears when the positive vector is normalized in (TN28.3).

The second theorem

> **Second fundamental theorem.** Assume no arbitrage. The market is complete (every payoff in $\mathbb{R}^n$ is attainable by some portfolio) if and only if the state-price vector, equivalently the martingale measure, is unique.

This one is a statement about the rank of D. Completeness says the rows of D span $\mathbb{R}^n$, that is, $\operatorname{rank} D = n$. The solutions of $S = D\psi$ form an affine set: one particular solution plus the null space of D acting on the right. That null space is trivial exactly when $\operatorname{rank} D = n$. So the solution is unique if and only if the market is complete, and the strict positivity carried over from the first theorem does not affect the count. ■

The economic reading is the useful one. When the measure is unique, every payoff has a replicating portfolio and its price is the cost of that portfolio: the price is a fact about the market. When it is not, the attainable payoffs are a proper subspace, and a claim outside it has a *range* of prices consistent with no arbitrage, one for each admissible measure.

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2 WHAT THE RANGE OF PRICES IS

The interval is not an abstraction; its endpoints are trades. For a payoff X not in the span, define

$$\overline{\pi}(X) = \min_{\theta} \left\{ \theta^\top S : \theta^\top D \geq X \right\}, \qquad \underline{\pi}(X) = \max_{\theta} \left\{ \theta^\top S : \theta^\top D \leq X \right\} \tag{TN28.6}$$

the cheapest portfolio that dominates the payoff in every state and the dearest that is dominated by it. Linear programming duality applied to (TN28.6), with the dual variables being exactly the state-price vectors, gives

$$\underline{\pi}(X) = \inf_{\psi} \ \psi^\top X, \qquad \overline{\pi}(X) = \sup_{\psi} \ \psi^\top X \tag{TN28.7}$$

the infimum and supremum taken over all strictly positive state-price vectors; neither is attained, because the optimum puts a zero price on some state, so for a payoff outside the span the arbitrage-free prices are the open interval between the sub-replication and super-replication costs, and its width measures how much of the payoff cannot be manufactured from what trades.

Example 2.1 What "a range of prices" actually means

One period, zero interest, a stock at 100 that can finish at 80, 100, or 130. Two instruments trade: the bond and the stock. Price an at-the-money call, whose payoffs are $(0, 0, 30)$.

A martingale measure is any (q_1, q_2, q_3) that is strictly positive, sums to one, and reprices the stock:

$$80q_1 + 100q_2 + 130q_3 = 100$$

Substituting $q_2 = 1 - q_1 - q_3$ collapses this to $q_1 = 1.5\,q_3$, so the whole family is described by one free number:

$$(q_1, q_2, q_3) = (1.5q_3,\ 1 - 2.5q_3,\ q_3), \qquad 0 < q_3 < 0.4$$

Every one of these is a legitimate arbitrage-free measure, and the call is worth $30q_3$, so its price ranges over

$$0 < c < 12$$

The theory does not narrow an at-the-money call to a neighborhood of some fair value; it says only that the price lies strictly between nothing and 12, on a stock worth 100. Any model that returns a single number (5.50, say) has chosen $q_3 = 0.18\overline{3}$ for reasons the market has not supplied.

The endpoints are the trades of (TN28.6). The cheapest portfolio that dominates the payoff in every state solves

$$\min\ 100a + b \quad \text{subject to} \quad 80a + b \geq 0,\ \ 100a + b \geq 0,\ \ 130a + b \geq 30$$

which binds in the outer two states at $a = 0.6$, $b = -48$, for a cost of **12**; the most expensive portfolio dominated by the payoff is the empty one, worth **0**.

Now add one instrument and the whole structure collapses to a point. Suppose a call struck at 100 trades at 5.50. That pins $q_3 = 5.50/30 = 0.18\overline{3}$, hence $q_1 = 0.275$ and $q_2 = 0.541\overline{6}$, and every other claim on this stock now has a unique price: a digital paying 1 at 130 is worth 0.1833, one paying 1 at 80 is worth 0.275, and no model input is required for either. Completing a market is not a mathematical operation but a market-structure fact: somebody had to quote the third instrument.

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That is the second theorem in miniature, and it explains the pattern of the book. Where a third instrument is quoted (vanilla options across strikes in Chapter 19), prices are pinned and models interpolate. Where it is not (forward volatility in Section 21.3, correlation in Section 31.12, recovery in Section 24.4), the interval stays open and what fills it is an assumption written in the notation of a price.

3 CONTINUOUS TIME

Both theorems were stated for finitely many states, and the rest of the book works in continuous time. *The first theorem is false there as stated*, and not for a technical reason about measurability.

The trouble is a doubling strategy. With unlimited credit a trader can compress an unbounded martingale betting scheme (double the stake after each loss, stop at the first gain) into a finite interval and finish with a certain profit on no initial investment. The construction needs no market imperfection and works in the Black–Scholes–Merton model as readily as anywhere else. It is an arbitrage in the sense of (TN28.1), and it exists because the strategies allowed were never restricted.

So the class of strategies has to be cut down before the theorem can be true. A strategy is **admissible** if its wealth process is bounded below by some constant $-a$; a doubling scheme is inadmissible because its interim losses are unbounded. With that restriction the theorem returns in a sharpened form, due to Delbaen and Schachermayer (1994):

> **First fundamental theorem, continuous time.** For a locally bounded semimartingale discounted price process, **no free lunch with vanishing risk** holds if and only if there exists an equivalent *local* martingale measure.

Two words changed and both matter.

No free lunch with vanishing risk (NFLVR) is stronger than the absence of arbitrage. It rules out not only arbitrages but sequences of admissible strategies whose downside shrinks to zero while a strictly positive profit survives in the limit: an approximate arbitrage that no single strategy realizes. Plain no-arbitrage is not enough to produce a measure in continuous time; NFLVR is exactly enough.

Local martingale is weaker than martingale. A local martingale is a process that is a martingale when stopped at each of an increasing sequence of times exhausting the horizon, and it need not be a martingale outright. The distinction bites here: if the discounted price of an asset is a strict local martingale (a local martingale that is not a true martingale), then

$$S_0 > E^{Q}\left[e^{-rT} S_T\right] \tag{TN28.8}$$

and the asset trades above the value of the payoff it delivers. This is what a *bubble* means in this literature, and it is a property of the model rather than of the market: put–call parity fails, and the Black–Scholes–Merton formula is no longer the price of a call. It is a reason to check that a calibrated model's discounted price is a true martingale, with expectation equal to today's price, rather than to assume it is.

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The second theorem survives the move intact. Completeness is still equivalent to uniqueness of the equivalent local martingale measure, and in a diffusion model driven by a Brownian filtration the machinery behind it is the martingale representation theorem: every square-integrable martingale on that filtration can be written as a stochastic integral against the Brownian motion. That representation is the formal content of the claim that a delta hedge exists, which is why the Black–Scholes–Merton model is complete and why models that add a risk no traded asset spans, such as stochastic volatility and jumps, are not.

4 WHERE INCOMPLETENESS SHOWS UP

The three cases of Chapter 28's Question 13 are (TN28.7) with more states.

Stochastic volatility is incomplete because volatility is not traded, so a second hedging instrument is missing and the loadings matrix of Chapter 28's Section 28.3 is not invertible. Section 21.10's parameter identification problem is the practical form of that incompleteness.

Jumps are incomplete because a jump of unknown size cannot be hedged by a finite set of instruments: the number of states reachable in an instant exceeds the number of assets available to span them.

Credit is incomplete for the reason Section 24.4 gives: recovery and default probability are not separately identifiable from spreads, so the model must supply what the market does not determine.

The theorems are therefore not reassurance but a diagnostic. When a price depends on a modeling choice, it is because the market is incomplete in a specific way, and identifying that way is more useful than reporting a calibration error.

REFERENCES

Harrison and Kreps (1979) established the discrete-time theorems, and Harrison and Pliska (1981) carried them into continuous time; both are demanding and are where to go for precise statements. Delbaen and Schachermayer (1994) is the source of the continuous-time result quoted above, and their later book collects the whole program. Duffie's *Dynamic Asset Pricing Theory* gives the finite-state proofs in the form used in Section 1, and Björk's textbook, which develops the measure theory it needs as it goes, is the place to start on the martingale representation argument.

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